EVERY TWO-COLORING OF THE PLANE CONTAINS A MONOCHROMATIC DEGENERATE TRIANGLE WITH SIDES 1, 3, AND 4

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Abstract. We prove that every two-coloring of the Euclidean plane contains three collinear points of one color with consecutive distances 1 and 3. Suppose, for a contradiction, that some two-coloring contains no such triple. We consider the triangular lattice $\Lambda = \mathbb{Z}[\omega]$ together with its two rotated copies $q\Lambda$ and $\bar{q}\Lambda$, where $q = \frac{8+5\omega}{7}$. Each copy shares a sublattice with Λ, so the colorings of the three lattices constrain each other. We show that, in one of its three directions, every row of a unit triangular lattice is colored in monochromatic runs of length exactly four, and that consecutive such rows are related by a fixed shift. Then any two points of the plane at distance 8 have the same color, which is impossible under our assumption.

1. Introduction

Let $\mathbb{E}^2$ denote the Euclidean plane. Suppose that we have a coloring of $\mathbb{E}^2$ using two colors. We are interested in three-point configurations that appear monochromatically in every such coloring.

Definition 1.1. The *triple* $T(a, b, c)$ is a triple of points forming the vertices of a triangle whose sides have lengths a, b, and c. This includes collinear configurations, which correspond to *degenerate* triangles, with one side equal to the sum of the other two.

The notation $T(a)$ is shorthand for $T(a, a, a)$, a triple of points forming an equilateral triangle of side length a.

Definition 1.2. A coloring *contains* $T(a, b, c)$ exactly when some congruent copy of it is monochromatic.

A foundational contribution to Euclidean Ramsey theory—which studies such questions, as well as more general ones—was made in the 1970s by Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus. In particular, they established in [EGM+73] that for an equilateral triangle with side a, there exists a two-coloring avoiding $T(a)$, made out of parallel stripes of alternating colors, each of width $\frac{a\sqrt{3}}{2}$. Later, Jelínek, Kynčl, Stolař, and Valla [JKS+09] also found non-trivial colorings avoiding $T(a)$. These remain the only triples known not to be contained in every two-coloring.

In 1975, Erdős et al. [EGM+75] identified several families of triples that are contained in every two-coloring of the Euclidean plane. These include configurations $T(a, b, c)$ such that:

- the ratio $\frac{a}{b}$ equals $2\sin\left(\frac{\theta}{2}\right)$ for some $\theta \in \{30°, 36°, 108°, 120°, 150°\}$[1];
- the triangle has an angle of 30° or 150°;
- the triangle has angles $(\alpha, 2\alpha, 180° - 3\alpha)$ for $0 < \alpha < 60°$, $\alpha \neq 45°$;
- the triangle has angles $(180° - \alpha, 180° - 2\alpha, 3\alpha - 180°)$ for $60° < \alpha < 90°$;
- $T(a, b, c) = T(a, 2a, 3a)$, a degenerate triangle;
- a, b, and c satisfy $c^2 = a^2 + 2b^2$, with $b < 2a$;
- a, b, and c satisfy $a^6 - 2a^4b^2 + a^2b^4 - 3a^2b^2c^2 + b^2c^4 = 0$;
- a, b, and c satisfy $a^4c^2 + b^4a^2 + c^4b^2 - 5a^2b^2c^2 = 0$;
- a, b, and c are the base, the leg, and the circumradius of an isosceles triangle.

In the following year, Shader [Sha76] added the triples $T(a, b, c)$ for which

- the triangle is right-angled;
- a, b, and c satisfy $a^2 + c^2 = 4b^2$, where $\frac{\sqrt{7}-1}{2}\, b \leq a \leq \frac{\sqrt{7}+1}{2}\, b$ (Shader's original range for a was narrower and was extended in the author's thesis [Yan25]).

[1] Erdős et al. erroneously mention the angle 72° instead of 36° in their paper.

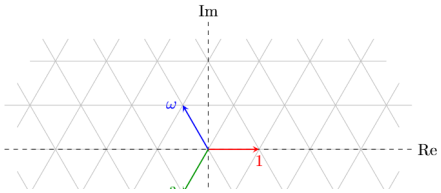
After these classical results, there was no progress on new triples contained in every two-coloring for over four decades, until Currier, Moore, and Yip [CMY24] proved it for $T(1, 1, 2)$, which was an inspiration to this research.

Theorem 1.3. *For every $a > 0$, every two-coloring of $\mathbb{R}^2$ contains $T(a, 3a, 4a)$.*

Theorem 1.3 is proved in Section 3. Several lemmas about runs are taken from the author's thesis [Yan25] (Section 2.2), as is the idea of overlapping a lattice with a rotated copy, proposed in its conclusion.

2. Preliminaries

2.1. **Notation and key triples.** The *Eisenstein integers* are the complex numbers $a + b\omega$ with $a, b \in \mathbb{Z}$, where $\omega = e^{\frac{2\pi i}{3}} = \frac{-1+i\sqrt{3}}{2}$ is a primitive cube root of unity; we write $\Lambda = \mathbb{Z}[\omega] = \mathbb{Z} + \mathbb{Z}\omega$ for the set of all of them. Then $\omega^2 = -1 - \omega$, $\bar{\omega} = \omega^2$, and $|a + b\omega|^2 = a^2 - ab + b^2$, and Λ, viewed as a subset of the plane, is the triangular lattice (Figure 1). A *unit triangular lattice* is a set $z_0 + u\Lambda$ with $|u| = 1$; its three *directions* are $u, u\omega, u\omega^2$, and its *rows* in a direction are the lines parallel to it.



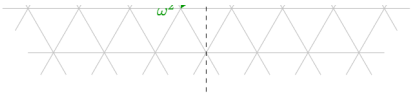


FIGURE 1. The lattice Λ is spanned by 1 and ω.

A two-coloring of the plane is *admissible* if it does not contain $T(1,3,4)$; we assume one is given and derive a contradiction.

Theorem 2.1 ([EGM+75])**.** *A two-coloring of the plane contains $T(a,b,c)$ if and only if it contains $T(a)$, $T(b)$, or $T(c)$.*

The proof in [EGM+75] applies equally to degenerate triangles. In particular, for $T(1,3,4)$ it means that a two-coloring of the plane contains $T(1,3,4)$ if and only if it contains $T(1)$, $T(3)$, or $T(4)$. Therefore, an admissible coloring of the plane contains none of $T(1)$, $T(3)$, and $T(4)$.

Definition 2.2. A *key triple* is a $T(1)$, $T(3)$, $T(4)$, or $T(1,3,4)$; in an admissible coloring, no key triple is monochromatic.

2.2. **Runs in a lattice.** We need three facts about a single lattice, all from the analysis of $T(1,3,4)$-free colorings of the triangular lattice in [Yan25, Section 3]; we include complete proofs for admissible colorings of the plane. Fix an admissible coloring and a unit triangular lattice L; no key triple in L is monochromatic. A *run* is a maximal set of consecutive points of one color on a row.

Lemma 2.3 (runs)**.**

(a) *Every run has at most four points.*
(b) *If a run has at least three points, then the runs next to it on both sides have the other color and at least three points.*